

Point-Level Refinable Semantic Mapping for Spatial Understanding

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Abstract—Open-vocabulary 3D semantic maps have emerged as a key spatial memory representation for long-horizon robot autonomy. However, existing object-centric maps are designed for static environments and rely on append-only point integration: newly observed points are accumulated while previously integrated points are rarely reevaluated, so perception errors accumulate and outdated points persist after objects move or disappear. Although recent dynamic mapping methods update object-level representations, they do not explicitly revise noisy or obsolete points within each object. We propose an online refinable semantic mapping framework that continuously refines object representations through point-level probabilistic confidence updating. Each object point maintains a confidence score updated online from multi-view temporal evidence: consistent observations reinforce reliable points, while inconsistent segmentation masks, depth measurements, and visibility cues suppress spurious and outdated points. Combined with tracking and re-identification that preserve instance identities under motion and relocation, our framework enables maps to be continuously refined rather than monotonically accumulated. Experiments show improved mapping accuracy over the previous state of the art by 3.46% on Replica and 2.55% on ScanNet, robustness in dynamic environments and under perception noise, and effective spatial memory for object navigation and embodied question answering with success rates of 82.4% and 76.7%.

I. INTRODUCTION

Open-vocabulary 3D semantic maps have emerged as a key spatial memory representation for robot autonomy [1] by grounding open-ended semantic concepts in 3D space [2]. They enable robots to localize task-relevant objects, reason about object relationships, and maintain persistent spatial context for embodied tasks, e.g., vision-language navigation [3] and mobile manipulation [4].

Recent open-vocabulary mapping systems [5], [6], [7] are designed for static scenes and rely on append-only point integration: newly observed points are continuously added to object instances, while previously integrated points are rarely revisited. Dynamic mapping methods [8], [9], [10] update object-level representations in changing scenes, yet they do not explicitly revise noisy or outdated points within each instance, so perception artifacts and stale observations remain embedded in the representation and degrade long-term mapping quality.

To address both dynamic environments and noisy perception, we propose an online refinable open-vocabulary semantic mapping framework that continuously refines object-centric maps at the point level. Each object point maintains a probabilistic confidence score updated online from multi-view temporal evidence: points consistently supported by

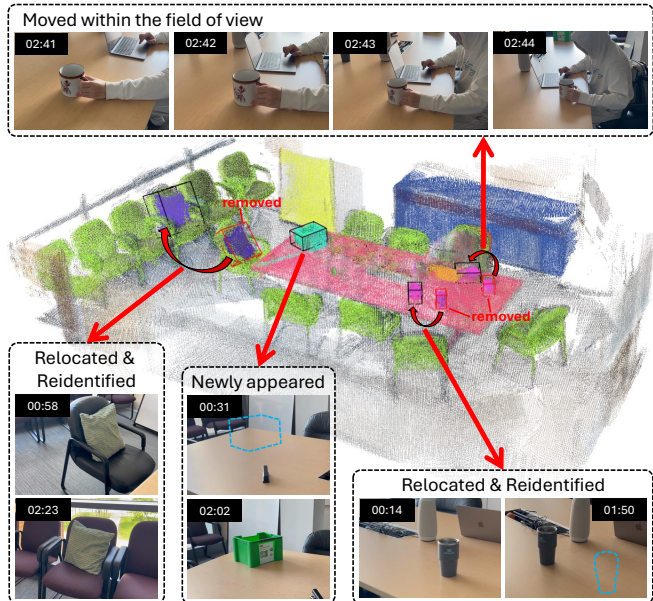


Fig. 1: **Dynamic open-vocabulary semantic mapping.** Our method maintains a refinable object-centric 3D semantic map through point-level probabilistic updates. It handles dynamic objects that newly appear, move within the robot’s view, or are relocated out of view and later re-identified.

new observations are reinforced, while those inconsistent with segmentation masks, depth measurements, or visibility cues are down-weighted and eventually removed. To preserve consistent identities, the framework combines 2D tracking for in-view motion with feature-based re-identification for out-of-view relocation, as shown in Fig. 1. We further integrate the map into a vision-language navigation (VLN) framework [3] and evaluate it on object navigation and embodied question answering (EQA) in real-world environments. Our contributions are:

- We propose an online refinable semantic mapping framework with point-level confidence updating, enabling object-centric semantic maps to be continuously refined rather than monotonically accumulated.
- We introduce a multi-view temporal update mechanism that reinforces reliable points while suppressing spurious and outdated points caused by perception errors and object motion.
- We integrate object tracking and feature-based re-identification to preserve consistent object identities, and demonstrate the map as an effective online spatial memory for VLN and EQA.

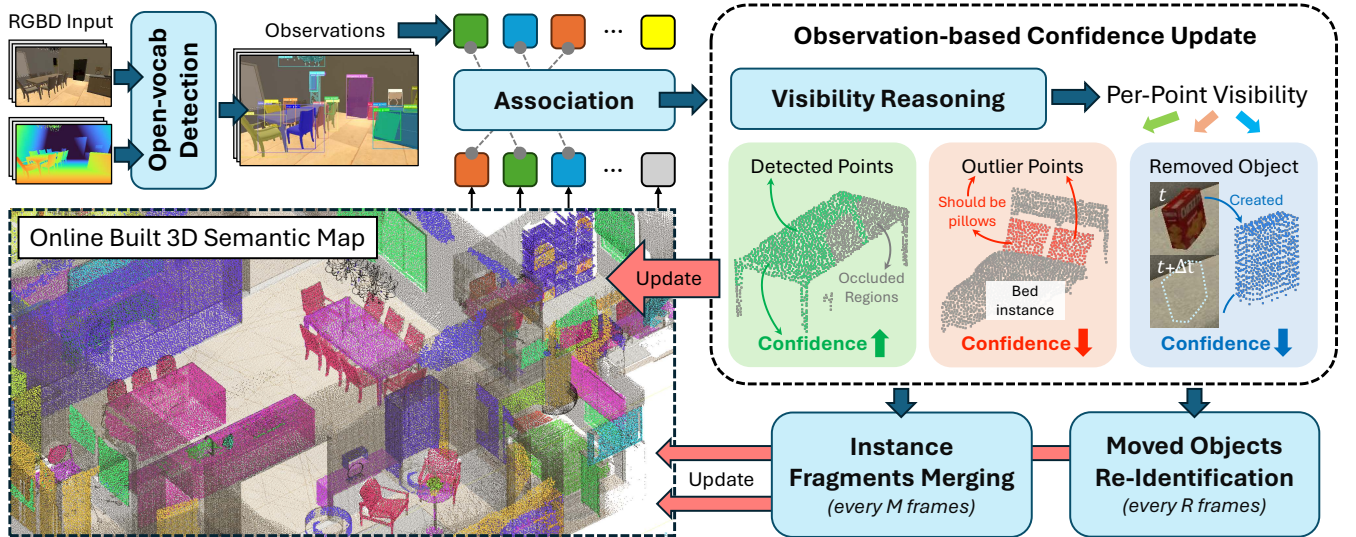


Fig. 2: **Overview of the proposed framework.** Given posed RGB-D observations, open-vocabulary detections are associated with object instances. For each association, visibility-based point confidence updates reinforce points supported by current detections (green), down-weight inconsistent outliers caused by earlier segmentation errors (red), and prune points belonging to removed objects. Unobserved points (gray) remain unchanged until more evidence is collected. Periodic fragment merging reduces duplicate instances, and re-identification preserves identities when objects are relocated and re-observed.

II. RELATED WORK

A. Open-Vocabulary Semantic Mapping

Semantic mapping augments geometric reconstruction with object labels, instance structure, and region-level semantics for navigation and manipulation [2]. Early metric-semantic systems rely on closed-set detectors [11], while recent vision-language foundation models [12], [13] enable open-vocabulary maps that can be queried with free-form language. Building on these advances, recent systems [5], [6], [7] integrate open-vocabulary maps with LLMs for embodied reasoning and long-horizon planning. These approaches demonstrate the value of structured semantic memory, but they assume fused object points remain valid once inserted, making erroneous or old points difficult to revise through later observations.

B. Dynamic Semantic Map Updating

Dynamic semantic mapping systems reason about environmental changes using geometric consistency and object relations. Khronos [8] removes outdated objects through ray-casting-based depth consistency checks when revisiting previously mapped areas. OpenIN [9] models carrier relationships between movable objects and their supporting objects, updating the relation graph through local observations. DualMap [10] combines a concrete local map with an abstract global map and re-selects candidate targets when a previous target is found to have moved. These methods enable maps to adapt to dynamic scenes, but they mainly perform object-level updates and lack mechanisms to revise noisy points introduced by detection, segmentation, or depth errors.

III. METHOD

Given an RGB-D stream, our goal is to build an online open-vocabulary 3D semantic map in which each observation both adds new object points and revises existing ones through detection and visibility cues, as illustrated in Fig. 2.

A. Probabilistic Map Representation

We represent the map as a set of object instances, where each instance stores its 3D points, per-point confidence scores, a label histogram, a bounding box, observation statistics, and CLIP [14] / DINO [15] descriptors:

$$\mathcal{M} = \{o_i : (\text{id}_i, \mathcal{P}_i, \mathcal{S}_i, \mathbf{h}_i, \text{bb}_i, \mathbf{f}_i^{\text{clip}}, \mathbf{f}_i^{\text{dino}}, n_i^{\text{obs}}, t_i^{\text{obs}})\} \quad (1)$$

The per-point score s_{ij} is the log-odds of its confidence probability c_{ij} ,

$$s_{ij} = \sigma^{-1}(c_{ij}) = \log \frac{c_{ij}}{1 - c_{ij}}, \quad (2)$$

so that new evidence is added directly to the existing score — equivalent to Bayesian probability updating but more computationally efficient [16].

B. Online Confidence Updating

Each incoming frame is treated as evidence for both map expansion and correction. Open-vocabulary detections with segmentation masks are predicted by vision foundation models:

$$\{\mathbf{m}_k, l_k, c_k, \mathbf{g}_k^{\text{clip}}, \mathbf{g}_k^{\text{dino}}\}_{k=1}^{n_t} = \text{Perception}(\mathbf{I}_t, \text{prompt}), \quad (3)$$

where $\mathbf{m}_k$, l_k , and c_k are the mask, semantic label, and confidence of the k -th detection, and $\mathbf{g}_k^{\text{clip}}$, $\mathbf{g}_k^{\text{dino}}$ are the CLIP/DINO features of the detected region. Each detection is lifted to a 3D point cloud in the world frame via depth back-projection and cleaned with DBSCAN [17] retaining

the largest connected component. Every map point is then projected into the current view and labeled *visible*, *missed*, or *occluded* by comparing its projected depth d_{ij} with the observed pixel depth D_t :

$$v_{ij} = \begin{cases} \text{visible,} & |d_{ij} - D_t(\mathbf{u}_{ij})| \leq \epsilon_d, \\ \text{missed,} & d_{ij} < D_t(\mathbf{u}_{ij}) - \epsilon_d, \\ \text{occluded,} & \text{otherwise,} \end{cases} \quad (4)$$

where $\mathbf{u}_{ij}$ is the projected pixel coordinate and ϵ_d is a small depth tolerance. Only points within a local distance window of the camera are processed, keeping the cost independent of the global map size. Detections are associated with existing instances using the sum of CLIP cosine similarity and 3D intersection-over-union (IoU). Unlike prior works [6], [10], the overlap is computed over the *visible* portion of the object rather than the full model, yielding more reliable matching under partial observability.

Confidence scores are then updated according to four cases. (1) *In-mask points* — visible points whose reprojections fall inside the matched mask — receive positive evidence and are increased by the log-odds of the detection confidence. (2) *Out-of-mask points* receive a fixed negative penalty, correcting over-fused regions. (3) If an instance is not associated with any detection but its image-space visibility ratio and footprint (computed via a morphological closing over the reprojected region) indicate it should be detectable, all its point scores are decreased, so that “ghost” instances created by erroneous detections are removed while occluded or weakly visible objects are spared. (4) Points classified as *missed*, which contradict the current depth observation along the same viewing rays, receive a strong penalty; this removes points left behind when objects are moved or removed. After each frame, points whose scores fall below a threshold are pruned, as are instances with too few points. This online updating preserves stable, consistently supported geometry while gradually eliminating artifacts caused by perception noise or object motion.

C. Fragment Merging and Identity Preservation

Object fragmentation, where disconnected partial observations of the same physical object are initialized as separate instances, is corrected by periodically merging instances whose CLIP similarity, DINO similarity, and 3D point-cloud overlap all exceed thresholds — a conservative rule that requires agreement in semantics, appearance, and geometry.

Stable instance identities are maintained under object motion. For *in-view* motion, the lightweight 2D tracker BoT-SORT [18] associates detections across frames, so a moving object is fused into the same instance rather than re-initialized; trailing points left at previous locations are removed by the confidence update within the next few frames. For *out-of-view* relocation, we store a compact record (instance ID, bounding box, visual features) with an importance score when an instance is removed by depth contradiction, in a bounded buffer. A removed record and a current instance are matched when CLIP similarity, DINO similarity, and bounding-box volume consistency all hold.

TABLE I: Open-vocabulary semantic mapping results on Replica and ScanNet.

Dataset	Method	mIoU	F-mIoU	mAcc
Replica	ConceptGraphs	0.1501	0.3858	0.3559
	HOV-SG	0.2050	0.4846	0.3835
	DualMap	0.2538	0.5207	0.4024
	Ours	0.2629	0.5530	0.4671
ScanNet	ConceptGraphs	0.0882	0.3077	0.3538
	HOV-SG	0.1333	0.3381	0.3714
	DualMap	0.1604	0.3288	0.3794
	Ours	0.1646	0.3405	0.3850

TABLE II: Computational and memory efficiency.

Dataset	Method	Avg Mem (MB)	Peak Mem (MB)	Avg TPF (s)
Replica	DualMap	3004.5	4668.6	0.260
	Ours	3056.9	3126.8	0.221
ScanNet	DualMap	2035.9	2873.9	0.146
	Ours	2982.2	2998.5	0.179

Ambiguous matches (e.g., a removed chair among several identical chairs) are skipped; a unique match observed consistently across three attempts restores the original instance ID. This process is independent of observation order: objects may disappear and be re-identified, or appear first and be re-identified after removal from their old location is confirmed.

D. Integration with a VLN Framework

We integrate the map into VL-Nav [3] as a compact symbolic-spatial memory for LLM-based agents. A Qwen2.5-VL 7B [19] coordinator interprets language instructions and queries the map through fuzzy semantic queries (CLIP similarity between LLM-generated query terms and stored instance features), which return top- K matching objects as JSON metadata. For object navigation, the planner navigates to a located target or triggers frontier-based exploration otherwise; for EQA, the LLM decides whether enough evidence has been gathered or further exploration is needed.

IV. EXPERIMENTS

We evaluate (i) mapping accuracy against representative open-vocabulary baselines, (ii) robustness to perception errors, (iii) handling of object motion and relocation, and (iv) deployment on a real robot for VLN tasks.

A. Mapping Accuracy and Efficiency

Following [10], we use YOLO-World [12] for detection, MobileSAM [20] for masks, FastSAM [21] for off-prompt objects, and MobileCLIP (70% image / 30% text) and DINOv3 features. We evaluate on Replica [22] and ScanNet [23] against ConceptGraphs [5], HOV-SG [6], and DualMap [10], using mIoU, F-mIoU, and mAcc over the final maps (background-dominant classes excluded). ConceptGraphs and HOV-SG are offline methods, while DualMap is online, like ours. For association and confidence updating we set $\text{Sim}_{\text{th}} = 1.2$, $\epsilon_d = 3 \text{ cm}$, $\Delta_{\text{outlier}} = 0.2$, $\Delta_{\text{-det}} = 0.12$, and $\Delta_{\text{missed}} = 0.85$.

As shown in Table I, our method outperforms all baselines on both datasets: compared with DualMap, it improves mIoU by 3.46% and 2.55%, F-mIoU by 5.84% and 3.44%, and

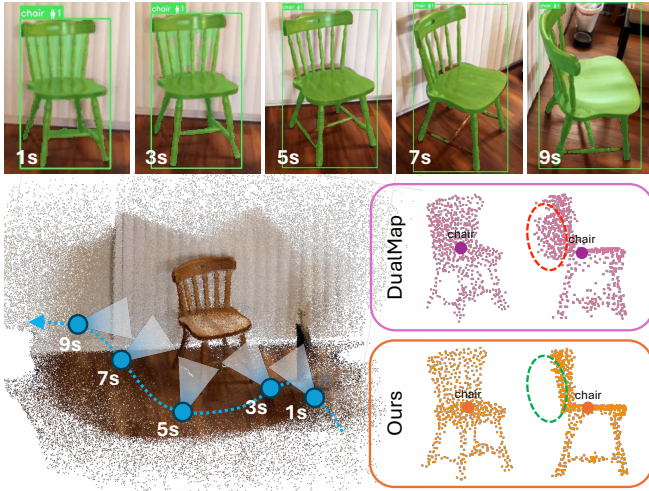


Fig. 3: **Handling inaccurate masks.** A wooden chair with thin, hollow structures: inaccurate masks fill the gaps between parts and introduce outlier points when fused to 3D. DualMap retains these artifacts, while our point-level confidence update suppresses inconsistent points across views and reconstructs cleaner geometry.

mAcc by 13.85% and 1.45% on Replica and ScanNet, respectively. Memory usage and per-frame time remain comparable to DualMap, as reported in Table II, owing to batched visibility reasoning and multiprocessing.

B. Robustness to Perception Errors

We conduct a perturbation test following [24], adding Gaussian noise $\mathcal{N}(0, \sigma = 3\text{px})$ to mask boundaries. Our mIoU drops 6.29% on Replica, versus 11.62% for DualMap: the point-level temporal evidence suppresses points introduced by segmentation errors. Beyond this quantitative test, our system mitigates four common error sources, as shown in Fig. 3: inaccurate masks for thin/hollow structures (a chair whose mask fills the gaps between back spindles), incorrect detections (a false “pillow” instance from a misdetected plastic bag is removed once later frames fail to support it), object fragmentation (a table observed as two fragments merges once geometry and appearance agree), and under-segmentation (points of a pillow fused into a chair are gradually removed once the pillow is correctly segmented).

C. Dynamic Object Handling

We evaluate tracking and re-identification in five HSSD scenes [25] with movable YCB objects [26]. For *in-view* motion, the map updates the same instance over time, preserving its ID while reflecting its new location, as shown in Fig. 4; over 32 moved objects we succeed in 27 cases (84.4%), with the remaining failures caused by objects becoming too small to detect when moved farther away. For *out-of-view* relocation, all 26 relocated objects are successfully removed from their original locations; 24 are detected at their new locations and 20 are re-identified with their original IDs (76.9%). Four detections are deliberately rejected to avoid false matches among visually similar objects.

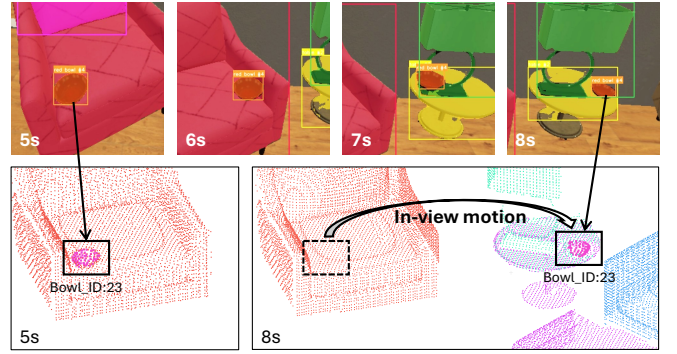


Fig. 4: **In-view object tracking.** A bowl moved from a chair to a table is tracked across frames; the semantic map updates the same instance over time, preserving its ID while reflecting its new location.

D. Real-World VLN Evaluation

We deploy the map on a Boston Dynamics Spot robot with RGB-D perception and LiDAR-based localization in three unseen environments (*Hallway*, *Office*, *Outdoor*). The map is built from scratch and updated online during execution, with no prior map or annotations. The LLM queries the map for target objects; navigation is planned when the target is localized and frontier-based exploration continues otherwise, while for EQA the agent explores until sufficient spatial evidence is gathered.

TABLE III: Real-world object navigation and embodied question answering.

Scenes	Object Navigation			EQA		
	Trials	SR	SPL	Trials	SR	SPL
Hallway	11	72.7	0.603	9	66.7	0.532
Office	12	91.7	0.790	12	83.3	0.714
Outdoor	11	81.8	0.685	9	77.8	0.641
All	34	82.4	0.696	30	76.7	0.638

As reported in Table III, the robot achieves an average success rate of 82.4% (SPL 0.696) over 34 object-navigation trials and 76.7% (SPL 0.638) over 30 EQA trials, demonstrating that the map reliably supports online object localization, spatial reasoning, and language-grounded behavior in unseen real environments.

V. CONCLUSION

We presented a dynamic open-vocabulary semantic mapping framework that turns object-centric maps from append-only accumulators into point-level probabilistic representations. By updating a log-odds confidence score per object point with multi-view evidence, the map reinforces stable points while suppressing perception artifacts and removing outdated points caused by object motion; tracking and re-identification preserve instance identities. Experiments on Replica, ScanNet, dynamic scenes, and a real robot validate improved accuracy, robustness, and the map’s effectiveness as online spatial memory for language-guided navigation and reasoning. Extending the map to discover novel categories autonomously is promising future work.

REFERENCES

- [1] Anonymous, “Imperative learning: A self-supervised neuro-symbolic learning framework for robot autonomy,” *International Journal of Robotics Research (IJRR)*, 2025.
- [2] D. Rotondi, F. Argenziano, S. Koch, N. Hughes, M. Buechner, J. Wald, L. R. Schmid, D. Nardi, A. Valada, L. Paull *et al.*, “3d scene graphs: Open challenges and future directions,” *arXiv preprint arXiv:2606.19383*, 2026.
- [3] Anonymous, “VL-Nav: A neuro-symbolic approach for reasoning-based vision-language navigation,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026, under review.
- [4] —, “Neuro-symbolic learning for long-horizon task planning under complex logical constraints,” *arXiv preprint*, 2026, under review.
- [5] Q. Gu, A. Kuwajerwala, S. Morin, K. M. Jatavallabhula, B. Sen, A. Agarwal, C. Rivera, W. Paul, K. Ellis, R. Chellappa *et al.*, “Conceptgraphs: Open-vocabulary 3d scene graphs for perception and planning,” in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 5021–5028.
- [6] A. Werby, C. Huang, M. Büchner, A. Valada, and W. Burgard, “Hierarchical Open-Vocabulary 3D Scene Graphs for Language-Grounded Robot Navigation,” in *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, July 2024.
- [7] D. Maggio, Y. Chang, N. Hughes, M. Trang, D. Griffith, C. Dougherty, E. Cristofalo, L. Schmid, and L. Carlone, “Clio: Real-time task-driven open-set 3d scene graphs,” *IEEE Robotics and Automation Letters*, vol. 9, no. 10, pp. 8921–8928, 2024.
- [8] L. Schmid, M. Abate, Y. Chang, and L. Carlone, “Khronos: A unified approach for spatio-temporal metric-semantic slam in dynamic environments,” in *Proc. of Robotics: Science and Systems (RSS)*, Delft, Netherlands, July 2024.
- [9] Y. Tang, M. Wang, Y. Deng, Z. Zheng, J. Deng, S. Zuo, and Y. Yue, “Openin: Open-vocabulary instance-oriented navigation in dynamic domestic environments,” *IEEE Robotics and Automation Letters*, vol. 10, no. 9, pp. 9256–9263, 2025.
- [10] J. Jiang, Y. Zhu, Z. Wu, and J. Song, “Dualmap: Online open-vocabulary semantic mapping for natural language navigation in dynamic changing scenes,” *IEEE Robotics and Automation Letters*, 2025.
- [11] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You only look once: Unified, real-time object detection,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 779–788.
- [12] T. Cheng, L. Song, Y. Ge, W. Liu, X. Wang, and Y. Shan, “Yolo-world: Real-time open-vocabulary object detection,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [13] N. Carion, L. Gustafson, Y.-T. Hu, S. Debnath, R. Hu, D. Suris, C. Ryali, K. V. Alwala, H. Khedr, A. Huang *et al.*, “Sam 3: Segment anything with concepts,” *arXiv preprint arXiv:2511.16719*, 2025.
- [14] P. K. A. Vasu, H. Pouransari, F. Faghri, R. Vemulapalli, and O. Tuzel, “Mobileclip: Fast image-text models through multi-modal reinforced training,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2024.
- [15] O. Siméoni, H. V. Vo, M. Seitzer, F. Baldassarre, M. Oquab, C. Jose, V. Khalidov, M. Szafraniec, S. Yi, M. Ramamonjisoa, F. Massa, D. Haziza, L. Wehrstedt, J. Wang, T. Darcet, T. Moutakanni, L. Sentana, C. Roberts, A. Vedaldi, J. Tolan, J. Brandt, C. Couprie, J. Mairal, H. Jégou, P. Labatut, and P. Bojanowski, “DINOv3,” *arXiv preprint arXiv:2508.10104*, 2025.
- [16] A. Hornung, K. M. Wurm, M. Bennewitz, C. Stachniss, and W. Burgard, “OctoMap: An efficient probabilistic 3D mapping framework based on octrees,” *Autonomous Robots*, 2013.
- [17] M. Ester, H.-P. Kriegel, J. Sander, X. Xu *et al.*, “A density-based algorithm for discovering clusters in large spatial databases with noise,” in *kdd*, vol. 96, no. 34, 1996, pp. 226–231.
- [18] N. Aharon, R. Orfaig, and B.-Z. Bobrovsky, “Bot-sort: Robust associations multi-pedestrian tracking,” *arXiv preprint arXiv:2206.14651*, 2022.
- [19] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, K. Dang, P. Wang, S. Wang, J. Tang *et al.*, “Qwen2.5-vl technical report,” *arXiv preprint arXiv:2502.13923*, 2025.
- [20] C. Zhang, D. Han, Y. Qiao, J. U. Kim, S.-H. Bae, S. Lee, and C. S. Hong, “Faster segment anything: Towards lightweight sam for mobile applications,” *arXiv preprint arXiv:2306.14289*, 2023.
- [21] X. Zhao, W. Ding, Y. An, Y. Du, T. Yu, M. Li, M. Tang, and J. Wang, “Fast segment anything,” *arXiv preprint arXiv:2306.12156*, 2023.
- [22] J. Straub, T. Whelan, L. Ma, Y. Chen, E. Wijmans, S. Green, J. J. Engel, R. Mur-Artal, C. Ren, S. Verma, A. Clarkson, M. Yan, B. Budge, Y. Yan, X. Pan, J. Yon, Y. Zou, K. Leon, N. Carter, J. Briales, T. Gillingham, E. Mueggler, L. Pesqueira, M. Savva, D. Batra, H. M. Strasdat, R. D. Nardi, M. Goesele, S. Lovegrove, and R. Newcombe, “The Replica dataset: A digital replica of indoor spaces,” *arXiv preprint arXiv:1906.05797*, 2019.
- [23] A. Dai, A. X. Chang, M. Savva, M. Halber, T. Funkhouser, and M. Nießner, “ScanNet: Richly-annotated 3d reconstructions of indoor scenes,” in *Proc. Computer Vision and Pattern Recognition (CVPR)*, IEEE, 2017.
- [24] N. Heller, J. Dean, and N. Papanikolopoulos, “Imperfect segmentation labels: How much do they matter?” in *International Workshop on Large-scale Annotation of Biomedical data and Expert Label Synthesis*. Springer, 2018, pp. 112–120.
- [25] M. Khanna*, Y. Mao*, H. Jiang, S. Haresh, B. Shacklett, D. Batra, A. Clegg, E. Undersander, A. X. Chang, and M. Savva, “Habitat Synthetic Scenes Dataset (HSSD-200): An Analysis of 3D Scene Scale and Realism Tradeoffs for ObjectGoal Navigation,” *arXiv preprint*, 2023.
- [26] B. Calli, A. Singh, J. Bruce, A. Walsman, K. Konolige, S. Srinivasa, P. Abbeel, and A. M. Dollar, “Yale-cmu-berkeley dataset for robotic manipulation research,” *The International Journal of Robotics Research*, vol. 36, no. 3, pp. 261–268, 2017.